Aalto University
School of Electrical
Engineering

ELEC-E7871

Advanced Topics in Human-Computer Interaction

Adaptive AI in Cooperative Cooking: Learning to Collaborate with Diverse Human Partners

Suryakant Sahoo
(suryakant.sahoo@aalto.fi)

Introduction

Human-AI collaboration is becoming vital across domains, but it considers human behaviour as ‘general’, while psychology & HCI researchers have shown that interactions based on **behaviour profiling** work better [1][2][3]. Overcooked is a great testbed for modelling human behaviours and observe how the **MARL-based** AI **learning curve** varies while collaboration with diverse types of simulated human partners.

Contribution

- **Modelling** Human Behaviour (Ranging from Adversarial to Helpful)
- AI **Training** Framework to adapt to diverse Human Behaviours
- **Insights** for Designing more flexible collaborative Human-AI system

Methods

ADAPTIVE AI IN **COOPERATIVE** COOKING
- Overcooked Game (5 x 5 grid)

ENVIRONMENT INTERACTION

HUMAN AGENT
- State Update
- Actions Update

AI AGENT
- State Update
- Actions Update

JOINT STATE
- Grid Visibility
- Object Positions
- Agent Positions

REWARD POINTS
- Metatask failed : -10
- Goodtask finished : 20
- Subtask finished : 50
- Correct delivery : 800
- Wrong delivery : -100
- Step penalty : -0.3
- Correct ingredient pickup : 15
- Sequence progress : 10
- Sequence completion : 50

AI AGENT POLICY

NEURAL NETWORK
- Input Layer (36)
- Hidden Layer (2 x 256)
- Output Layer (6)

LEARNING MECHANISM
Proximal Policy Optimization (PPO)

EXPERIENCE COLLECTION
- Past State History
- State Transition
- Advantage Estimation
- Returns Calculations

POLICY OPTIMIZATION
- Value Function Loss
- Entropy Regularization

POLICY UPDATE
- Different Policies emerge based on Human Behaviour type

HUMAN AGENT BEHAVIORS

STATIONARY
- No Movement

RANDOM
- Unpredictable

OBSTRUCTING
- Blocks AI Path (60%)

CLUMSY
- Makes Mistakes (60%)

SELFISH
- Resource Hoarding

HELPFUL
- Overly Helpful intent

LEARNED
- Learns Together with AI

BEHAVIOUR-SPECIFIC ADAPTATION OUTCOMES of AI

- Solo Task Completion
- Efficient Pathfinding

- Adaptability & Resilience
- Opportunistic Assistance

- Evasion & Resilience
- Alternative Paths

- Error Recovery
- Object Retrieval

- Resource Negotiation
- Task Division

- Task Division
- Coordination

- Complementary Actions
- Coordination

Figure 1 : Overview of the Multi Agent Reinforcement Learning based experiment for Human-AI collaboration

Results & Discussion

Based on the observed learning curves, we can clearly see that the adaptive AI agent changes its strategies depending on different human behaviour patterns. Some behaviours take longer to learn from, while others are easier to interpret and adapt to. Learning curves show **faster convergence when partner behaviour is consistent** (e.g., Stationary), but **better generalization** when the agent is trained with a **diverse set of behaviours**.

When both agents were learning from each other, training took about **50% longer** (for 200 iterations), yet convergence was the **second best overall**.

A noteworthy result emerged with the **‘Selfish’ and ‘Overly Helpful’** human behaviours: the AI’s performance was surprisingly similar in both cases, despite the human intentions being very different. This may be due to a lack of **explicit communication or planning**, or because the AI lack **role delegation awareness;** it is robust but not optimized for synergy.

Limitations of the current setup include: a small 5×5 grid, limited items to track (only 3), a simple task (tomato-salad delivery), a short training run (n = 200), and was tested with simulated human behavioural agents only.

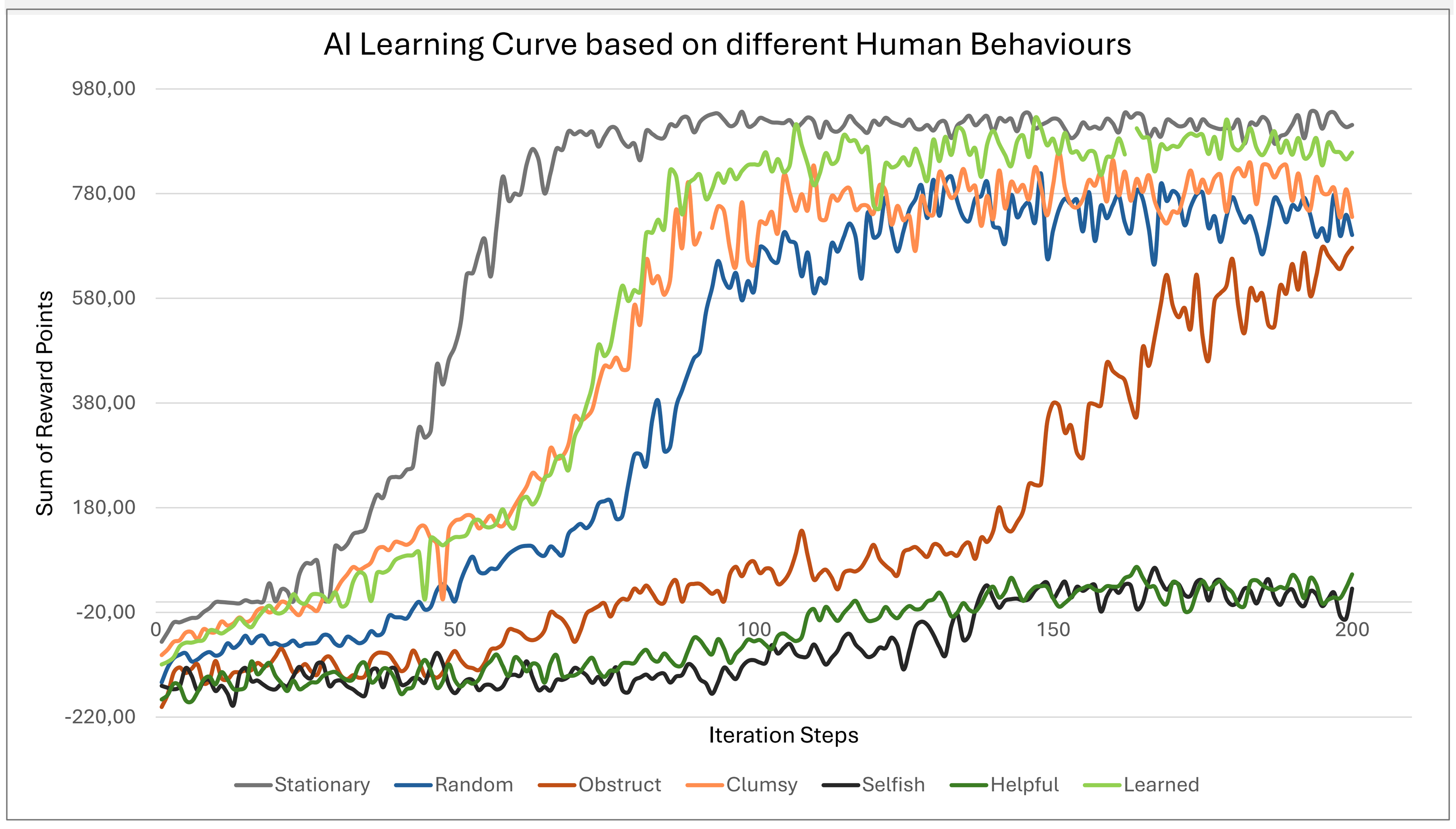

Figure 2 : Adaptive AI’s Learning Curve comparison for diverse simulated human behaviours

Conclusion

1. Training with diverse human behaviours builds robust but **not** necessarily cooperative AI
2. **Helpful ≠ Efficient**, unless coordination is learned. Coordination is key, not just good intentions. **Mutual learning** (AI & human) can lead to emergent cooperation
3. **Future Work :** Test with **real human participants** for behavioural grounding. Integrate **role prediction** or **cooperative planning modules.** Develop **collaboration scores** to evaluate each interaction

References

1. **Carroll, M., et al. (2019).** *On the Utility of Learning about Humans for Human-AI Coordination*. NeurIPS.
2. **Macrae, C. N., & Bodenhausen, G. V. (2000).** *Social Cognition: Thinking Categorically About Others.* Annual Review of Psychology.
3. **Wang, J., et al. (2020).** *Too Many Cooks: Coordinating Multi-Agent Collaboration Through Inverse Planning.* CogSci.